JASON ZHI LIANG, Ph.D.

San Francisco, CA

Contact: jasonzliang@utexas.edu | (510) 378-0709

LinkedIn: linkedin.com/in/jason-liang-3189b249

Scholar: scholar.google.com/citations?user=h8yYr7oAAAAJ

Github: github.com/jasonzliang

Research Summary

Principal Research Scientist dedicated to achieving Artificial General Intelligence (AGI) through the lens of nature-inspired computation and scalable reasoning. I specialize in the intersection of Evolutionary Optimization, Complexity Science, and Inference-Time Compute, moving beyond static scaling laws toward open-ended learning and recursive self-improvement.

My expertise lies in building agentic systems for deliberative System 2 reasoning and scaling population-based optimization for massive distributed compute. I designed and implemented Caesar, an autonomous Deep Research agent (analog of OpenAI Deep Research and Gemini Deep Research) that utilizes adversarial self-refinement and generative merging to achieve state-of-the-art creative synthesis. As the inventor of the CoDeepNEAT algorithm and lead author on 5 issued U.S. patents, I am eager to apply my experience in evolutionary, population-based optimization and architecture search to improve the next generation of AI agents and reasoning models.

Technical Skills and Expertise

- **Core Research:** Inference-Time Compute Scaling, Deep Research Agents, Agentic Reasoning (System 2), Associative Synthesis, Neural Architecture Search, Evolutionary Population-Based Optimization, Neuroevolution (Evolutionary Optimization for Deep Learning), Meta-Learning, Recursive Self-Improvement
- **Agent Infrastructure:** Multi-Agent Orchestration, Graph-Augmented RAG, Stateful Agentic Memory, FSM-Based Agent Coordination, Automated Evaluation (LLM-as-a-Judge), Evolutionary Prompt Optimization
- **Languages & Tooling:** Python (Advanced), PyTorch / Keras / TensorFlow, Hugging Face, Frontier LLM APIs, LlamaIndex, Multi-Agent Frameworks (AutoGen/MetaGPT), Automated Web Parsing

Professional Experience

Cognizant AI Lab | Principal Research Scientist

San Francisco, CA | Dec 2018 – Present | Lead Researcher driving the roadmap for the lab's AI Scientist project; team of 5 researchers.

1. Caesar: Autonomous Reasoning & Knowledge Synthesis * **Automated R&D via Inference-Time Self-Improvement:** Designed and implemented "Caesar," an autonomous Deep Research agent (analog of OpenAI Deep Research and Gemini Deep Research). Scaled test-time compute using an adversarial refinement loop that actively critiques internal drafts, generates orthogonal queries to target weaknesses, and consolidates findings via a generative merge. **Outperforms Gemini Deep Research, OpenAI Deep Research, and Claude Research Mode by 13–23% on creative synthesis benchmarks (Cliff's δ ≥ 0.76 across all output formats; corroborated by 23-person human eval).** *

Graph-Based Associative Reasoning: Replaced static RAG retrieval with a dynamic "Perceive-Think-Act" loop backed by a self-organizing knowledge graph, enabling discovery of non-obvious cross-disciplinary connections beyond the reach of linear retrieval agents. **Ablation: removing the knowledge graph and replacing it with flat top-T web search causes a large degradation (Cliff's δ = 0.52), confirming the graph drives a substantial share of Caesar's advantage.** * **Open-Ended Exploration Policy:** Designed an active information-foraging policy that detects stagnation in memory and graph structures, autonomously executing strategic backtracking to maximize information gain over long horizons. **Ablation: reducing exploration budget from 1000 to 250 iterations causes a −3.08 total-score decline (Cliff's δ = 0.92, large effect), validating that deep topological traversal is essential for creative synthesis.**

2. Multi-Agent Systems & Large Scale Evolutionary Optimization

- **Multi-Agent Code Generation:** Developed a **Finite State Machine (FSM)** framework to coordinate teams of specialized LLM agents. Applied evolutionary algorithms to optimize the *composition* of these teams, achieving competitive results on complex reasoning benchmarks like SciCode and HumanEval+.
- **Evolutionary Population-Based Training (EPBT):** Pioneered a population-based approach that evolves loss functions and hyperparameters jointly with weights via population-based weight sharing. **Outperforms the PBT baseline by 1.3pp on CIFAR-10 ResNet-32 (92.79% vs 91.53%) while exploring 520 candidate loss functions at 12.5× lower search cost than independent training** (GECCO 2021; U.S. patent pending).
- **Code Repository Analysis:** Created "Code Archaeologist," an agentic system capable of reasoning over repository histories (via Git/LFS) to detect architectural patterns and technical debt.

Sentient Technologies | Research Scientist

San Francisco, CA | June 2017 – Nov 2018

- **Massive-Scale Evolutionary Architecture Search:** Invented and scaled **CoDeepNEAT** (evolutionary neural architecture search) to clusters of hundreds of GPUs. Validated that population-based search scales with parallel compute — a direct precursor to today's PBT and inference-time scaling paradigms; established asynchronous evolutionary optimization at scale and forms the basis for 5 issued U.S. patents.
- **Multi-Task Optimization:** Achieved **88% test accuracy on the Omniglot 50-task multitask benchmark (+21pp over prior SOTA Soft Ordering at 67%)** by evolving modular topologies with cross-task weight sharing.
- **Production Deployment:** Collaborated with Sentient's product team to productionize CoDeepNEAT for image captioning.

The University of Texas at Austin | Research Assistant / Ph.D. Candidate

Austin, TX | Aug 2013 – Dec 2018

- **Solving Sparse Reward Problems:** Developed **MEA (Meta-Evolutionary Algorithm)**, a bilevel optimization framework. Successfully applied this to complex continuous control tasks (e.g., helicopter hovering), solving non-differentiable control problems where standard Gradient-based RL struggled to converge.

Selected Publications & Patents

Full publication list available on Google Scholar

Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis (2026) *Jason Liang, Elliot Meyerson, Risto Miikkulainen*, Preprint, arXiv:2604.20855. Under review at NeurIPS 2026.

- Autonomous R&D agent that achieves state-of-the-art knowledge synthesis by scaling inference-time compute through an adversarial refinement loop and a self-organizing knowledge graph.

Self-Transcendence: Achieving AGI via Chaotic Dynamics and Thermodynamic Attractors (2025) *Jason Liang*, TechRxiv Preprint.

- Position paper with framework for AGI based on non-equilibrium thermodynamics and self-organizing systems, moving beyond static objective functions.

Asynchronous Evolution of Deep Neural Network Architectures (2024) *Jason Liang, Hormoz Shahrzad, Risto Miikkulainen*, Applied Soft Computing, Vol. 152.

- Techniques for asynchronous population-based optimization at scale, essential for efficient training on massive distributed clusters.

Regularized Evolutionary Population-Based Training (2021) *Jason Liang, Santiago Gonzalez, Hormoz Shahrzad, and Risto Miikkulainen*, GECCO 2021.

- Method to dynamically optimize loss functions during training, improving convergence speed and generalization.

Evolutionary Neural AutoML for Deep Learning (2019) *Jason Liang, Elliot Meyerson, Babak Hodjat, et al.*, GECCO 2019.

- Evolutionary optimization of novel neural network architectures for multi-tasking learning.

Evolutionary Architecture Search for Deep Multitask Networks (2018) *Jason Liang, Elliot Meyerson, and Risto Miikkulainen*, GECCO 2018.

- Novel evolutionary algorithm (CoDeepNEAT) that automates the discovery of deep neural architectures, outperforming human-designed baselines.

Evolving Deep Neural Networks (2018) *Risto Miikkulainen, Jason Liang, Elliot Meyerson, et al.*, Artificial Intelligence in the Age of Neural Networks and Brain Computing.

- Foundational text establishing the synergy between Evolutionary Strategies (ES) and Deep Learning as a viable path toward scalable AI.

Evolutionary Bilevel Optimization for Complex Control Tasks (2015) *Jason Zhi Liang, Risto Miikkulainen*, GECCO 2015.

- Solving complex control tasks using evolutionary bilevel optimization, handling sparse reward signals where traditional RL failed.

Patents (Issued, U.S.):

- **US Patent 12,282,845:** Multi-objective Coevolution of Deep Neural Network Architectures (2025).
- **US Patent 11,507,844:** Asynchronous Evaluation Strategy for Evolution of Deep Neural Networks (2022).
- **US Patent 11,250,328:** Cooperative Evolution of Deep Neural Network Structures (2022).

- **US Patent 11,030,529:** Evolution of Architectures for Multitask Neural Networks (2021).
- **US Patent 11,003,994:** Evolutionary Architectures for Evolution of Deep Neural Networks (2021).

Patents (Pending, U.S.):

- **App. 17/389,961:** Regularized Evolutionary Population-Based Training (2023).
- **App. 17/064,706:** Sharing Meta-Learning Methods Among Multiple Private Data Sets (2022).

Professional Activities

- **ICML Silver Reviewer** — International Conference on Machine Learning (ICML) | 2025
- **Reviewer** — Genetic and Evolutionary Computation Conference (GECCO) | 2019 – Present

Education

Ph.D. in Computer Science (Neuroevolution & Deep Learning) *The University of Texas at Austin* | Sep 2013 – Dec 2018 *Advisor: Risto Miikkulainen*

B.S. in Electrical Engineering and Computer Science *University of California, Berkeley* | Sep 2009 – May 2013